Calculator and Logger

1.0

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1 Calculator and Logger Project 1
1.1 Introduction . . . 1
1.2 Features . . . 1
1.3 Project Files . . . 1

2 Todo List 3

3 Class Index 5
3.1 Class List . . . 5

4 File Index 7
4.1 File List . . . 7

5 Class Documentation 9
5.1 CalculationEngine Class Reference . . . 9
5.1.1 Detailed Description . . . 9
5.1.2 Constructor & Destructor Documentation . . . 10
5.1.2.1 CalculationEngine() . . . 10
5.1.3 Member Function Documentation . . . 10
5.1.3.1 add() . . . 10
5.1.3.2 multiply() . . . 10
5.1.3.3 power() . . . 11
5.1.3.4 subtract() . . . 11
5.1.4 Member Data Documentation . . . 11
5.1.4.1 logger . . . 11
5.2 Logger Class Reference . . . 12
5.2.1 Detailed Description . . . 12
5.2.2 Constructor & Destructor Documentation . . . 12
5.2.2.1 Logger() . . . 12
5.2.3 Member Function Documentation . . . 13
5.2.3.1 log() . . . 13
5.2.4 Member Data Documentation . . . 13
5.2.4.1 currentLevel . . . 13

6 File Documentation 15
6.1 CalculationEngine.h File Reference . . . 15
6.1.1 Detailed Description . . . 15
6.2 CalculationEngine.h . . . 15
6.3 Logger.h File Reference . . . 16
6.3.1 Detailed Description . . . 16
6.3.2 Enumeration Type Documentation . . . 17
6.3.2.1 LogLevel . . . 17
6.4 Logger.h . . . 17
6.5 main.cpp File Reference . . . 18

6.5.1 Detailed Description 18
6.5.2 Function Documentation 18
6.5.2.1 main() 18

Chapter 1

Calculator and Logger Project

1.1 Introduction

This project demonstrates a simple calculator application with a logging facility.

1.2 Features

- Addition
- Subtraction
- Multiplication
- Power calculation
- Logging with INFO, WARNING and ERROR levels

1.3 Project Files

- Logger.h: Defines the Logger class and LogLevel enumeration.
- CalculationEngine.h: Defines the mathematical operations.
- main.cpp: Contains the application entry point.

Chapter 2

Todo List

Class CalculationEngine

Add input validation for invalid mathematical operations.

Chapter 3

Class Index

3.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

CalculationEngine
: Performs basic mathematical operations 9

Logger
: Handles logging messages at different severity levels 12

Chapter 4

File Index

4.1 File List

Here is a list of all files with brief descriptions:

CalculationEngine.h
: Contains the CalculationEngine class 15

Logger.h
: Contains the Logger class for displaying log messages 16

main.cpp
: Contains the main function of the calculator application 18

Chapter 5

Class Documentation

5.1 CalculationEngine Class Reference

Performs basic mathematical operations.

`#include <CalculationEngine.h>`

Public Member Functions

- CalculationEngine (Logger &logger)
  Creates a CalculationEngine object.
- double add (double a, double b)
  Adds two numbers.
- double subtract (double a, double b)
  Subtracts the second number from the first.
- double multiply (double a, double b)
  Multiplies two numbers.
- double power (double base, double exponent)
  Calculates the power of a number.

Private Attributes

- Logger & logger

5.1.1 Detailed Description

Performs basic mathematical operations.

This class provides arithmetic operations such as addition, subtraction, multiplication and exponentiation. It uses the Logger class to display information about calculations.

Todo Add input validation for invalid mathematical operations.

5.1.2 Constructor & Destructor Documentation

5.1.2.1 CalculationEngine()

```
CalculationEngine::CalculationEngine (
            Logger & logger)  [inline]
```

Creates a CalculationEngine object.

Parameters

logger	Reference to the Logger object.

5.1.3 Member Function Documentation

5.1.3.1 add()

```
double CalculationEngine::add (
            double a,
            double b)  [inline]
```

Adds two numbers.

Parameters

a	First number.
b	Second number.

Returns

The sum of the two numbers.

5.1.3.2 multiply()

```
double CalculationEngine::multiply (
            double a,
            double b)  [inline]
```

Multiplies two numbers.

Parameters

a	First number.
b	Second number.

Returns

The product of the two numbers.

5.1.3.3 power()

```
double CalculationEngine::power (
            double base,
            double exponent)  [inline]
```

Calculates the power of a number.

Parameters

base	The base value.
exponent	The exponent value.

Returns

The result of raising base to exponent.

5.1.3.4 subtract()

```
double CalculationEngine::subtract (
            double a,
            double b)  [inline]
```

Subtracts the second number from the first.

Parameters

a	First number.
b	Second number.

Returns

The difference between the two numbers.

5.1.4 Member Data Documentation

5.1.4.1 logger

```
Logger& CalculationEngine::logger  [private]
```

The documentation for this class was generated from the following file:

- CalculationEngine.h

5.2 Logger Class Reference

Handles logging messages at different severity levels.

`#include <Logger.h>`

Public Member Functions

- Logger (LogLevel level=LogLevel::INFO)
  Constructs a Logger object.
- void log (LogLevel level, const std::string &message)
  Displays a log message.

Private Attributes

- LogLevel currentLevel

5.2.1 Detailed Description

Handles logging messages at different severity levels.

5.2.2 Constructor & Destructor Documentation

5.2.2.1 Logger()

```
Logger::Logger (
            LogLevel level = LogLevel::INFO)  [inline]
```

Constructs a Logger object.

Parameters

level	The minimum log level to display.

5.2.3 Member Function Documentation

5.2.3.1 log()

```
void Logger::log (
            LogLevel level,
            const std::string & message)  [inline]
```

Displays a log message.

Parameters

level	The severity level of the message.
message	The message to be displayed.

5.2.4 Member Data Documentation

5.2.4.1 currentLevel

```
LogLevel Logger::currentLevel  [private]
```

The documentation for this class was generated from the following file:

- Logger.h

Chapter 6

File Documentation

6.1 CalculationEngine.h File Reference

Contains the CalculationEngine class.

```
#include "Logger.h"
#include <cmath>
```

Classes

- class CalculationEngine

  Performs basic mathematical operations.

6.1.1 Detailed Description

Contains the CalculationEngine class.

6.2 CalculationEngine.h

Go to the documentation of this file.

```
00001
00005
00006 #ifndef CALCULATION_ENGINE_H
00007 #define CALCULATION_ENGINE_H
00008
00009 #include "Logger.h"
00010 #include <cmath>
00011
00022 class CalculationEngine
00023 {
00024 public:
00029     CalculationEngine(Logger& logger) : logger(logger) {}
00030
00037     double add(double a, double b)
00038     {
00039         double result = a + b;
00040         logger.log(LogLevel::INFO, "Addition performed");
00041         return result;
00042     }
00043
```

```
00050     double subtract(double a, double b)
00051     {
00052         double result = a - b;
00053         logger.log(LogLevel::INFO, "Subtraction performed");
00054         return result;
00055     }
00056
00063     double multiply(double a, double b)
00064     {
00065         double result = a * b;
00066         logger.log(LogLevel::INFO, "Multiplication performed");
00067         return result;
00068     }
00069
00076     double power(double base, double exponent)
00077     {
00078         double result = std::pow(base, exponent);
00079         logger.log(LogLevel::INFO, "Power calculation performed");
00080         return result;
00081     }
00082
00083 private:
00084     Logger& logger;
00085 };
00086
00087 #endif
```

6.3 Logger.h File Reference

Contains the Logger class for displaying log messages.

```
#include <iostream>
#include <string>
```

Classes

- class Logger

  Handles logging messages at different severity levels.

Enumerations

- enum class LogLevel { INFO , WARNING , ERROR }

  Represents the severity level of a log message.

6.3.1 Detailed Description

Contains the Logger class for displaying log messages.

6.3.2 Enumeration Type Documentation

6.3.2.1 LogLevel

```
enum class LogLevel  [strong]
```

Represents the severity level of a log message.

Enumerator

INFO	General information messages.
WARNING	Messages indicating a possible issue.
ERROR	Messages indicating an error.

6.4 Logger.h

Go to the documentation of this file.

```
00001
00005
00006 #ifndef LOGGER_H
00007 #define LOGGER_H
00008
00009 #include <iostream>
00010 #include <string>
00011
00015 enum class LogLevel
00016 {
00017     INFO,
00018     WARNING,
00019     ERROR
00020 };
00021
00026 class Logger
00027 {
00028 public:
00033     Logger(LogLevel level = LogLevel::INFO)
00034         : currentLevel(level) {}
00035
00041     void log(LogLevel level, const std::string& message)
00042     {
00043         if (static_cast<int>(level) >= static_cast<int>(currentLevel))
00044         {
00045             std::cout « "[";
00046
00047             switch (level)
00048             {
00049                 case LogLevel::INFO:
00050                     std::cout « "INFO";
00051                     break;
00052                 case LogLevel::WARNING:
00053                     std::cout « "WARNING";
00054                     break;
00055                 case LogLevel::ERROR:
00056                     std::cout « "ERROR";
00057                     break;
00058             }
00059
00060             std::cout « "] " « message « std::endl;
00061         }
00062     }
00063
00064 private:
00065     LogLevel currentLevel;
00066 };
00067
00068 #endif
```

6.5 main.cpp File Reference

Contains the main function of the calculator application.

```
#include "Logger.h"
#include "CalculationEngine.h"
```

Functions

- int main ()

6.5.1 Detailed Description

Contains the main function of the calculator application.

6.5.2 Function Documentation

6.5.2.1 main()

```
int main ()
```